

Addressing Class Imbalance in Cross-Domain Mosquito Bioacoustic Species Classification via Multi-Task Learning

Meiting Lyu¹, Ruicheng Liu¹, Xu Shen², Zongye Chen¹, Jiatong Wu¹, Ziyu Wang¹, Wei Liu¹, Gongping Huang¹

¹ Electronic Information School, Wuhan University, Wuhan, Hubei 430072, China

² Waseda University, Japan

Abstract—The Cross-Domain Mosquito Species Classification (CD-MSC) task focuses on mosquito species recognition under domain shift caused by variations across devices, environments, and collection protocols. Models trained on multiple source domains may perform poorly when evaluated on recordings collected in different settings. To address these challenges, we propose a robust classification framework that integrates diverse data augmentation, domain-invariant representation learning, and ensemble inference. Specifically, we employ multiple data augmentation strategies to enhance the diversity of training samples. To learn cross-domain features, we insert MixStyle modules into a DenseNet backbone to dynamically perturb feature statistics. Furthermore, the feature embedding is optimized using four complementary losses to mitigate cross-domain generalization issues. Finally, we adopt an ensemble inference strategy that leverages the strengths of different network architectures and training configurations for specific species. Experiments on the test set of the development dataset show that the proposed system achieves 0.4016 balanced accuracy on unseen domains, substantially outperforming the baseline of 0.1751.

Index Terms—mosquito species classification, class imbalance, domain-adversarial training, multi-task learning, bioacoustics

1. INTRODUCTION

Mosquito-borne diseases continue to pose a major threat to global public health, making efficient and accurate mosquito species identification essential for effective surveillance and control [1] [2]. Recently, deep learning techniques have also been applied to mosquito monitoring. Among these methods, computer vision approaches that identify mosquito species based on visual appearance have achieved considerable success [1]. However, vision-based methods are limited by image quality and capture conditions, making them inconvenient for application in field monitoring scenarios. Meanwhile, audio-based mosquito monitoring has emerged as a promising alternative [3] [4]. Audio-based monitoring does not require high-resolution imaging and is easy to deploy at low cost over large areas [5]. BioDCASE Mosquito Species Classification (MSC) aims to identify the species of a mosquito from its flight tone recordings (wingbeat sounds) [6] [7].

Regardless of the modality used, however, both vision- and audio-based approaches face the class imbalance of datasets across mosquito species, as some species are rarely captured or are only active during certain seasons [1] [6]. Although flight tone monitoring offers a low-cost deployment advantage [8], mosquito flight tones are narrow-band and are often recorded at a low signal-to-noise ratio and further vary across recording devices, geographical regions, and acoustic environments, making it difficult for models to learn species-relevant acoustic cues [4] [9]. This cross-domain issue is the central bottleneck for practical acoustic mosquito species classification [10].

The BioDCASE 2026 Cross-Domain Mosquito Species Classification (CD-MSC) challenge addresses this gap by providing a reliable development dataset and split, comprising nine mosquito species collected across five recording domains [6]. The challenge evaluation protocol reports balanced accuracy separately on seen domains

(BA_{seen}) and unseen domains (BA_{unseen}), allowing researchers to examine how strongly performance degrades in new domains and whether models retain species-relevant cues or instead rely on domain-specific cues [6]. By standardizing the dataset, split, and evaluation metrics, this benchmark provides the acoustic MSC community with a unified and reproducible reference for developing and comparing domain-robust classification methods.

In this paper, we propose a robust classification framework for the CD-MSC task that jointly addresses cross-domain generalization and severe class imbalance. Our approach incorporates the following contributions:

- **Diverse Data Augmentation:** We employ a combination of data augmentation strategies to enhance the diversity of training samples, mitigating overfitting to species with limited recordings and to site-specific recording artifacts.
- **Domain-Invariant Representation Learning:** We insert MixStyle modules into a DenseNet backbone to dynamically perturb feature statistics during training, encouraging the encoder to learn domain-invariant, species-relevant acoustic cues rather than relying on domain-specific recording characteristics.
- **Multi-Loss Feature Optimization:** We optimize the feature embedding with four complementary losses that jointly enforce species discriminability and domain robustness, mitigating cross-domain generalization issues beyond what a single classification objective can achieve.
- **Ensemble Inference:** We adopt an ensemble inference strategy that leverages the complementary strengths of different network architectures and training configurations specialized for specific species, improving robustness against unseen domain shifts.

Experiments on the test set of the development dataset show that the proposed system achieves a balanced accuracy of 0.4016 on unseen domains, substantially outperforming the official CD-MSC baseline of 0.1751 [6] and verifying the effectiveness of our robust classification framework.

2. METHODOLOGY

This section presents the overall framework, including feature extraction, augmentation, model architecture, and the multi-task training objective, as illustrated in Fig. 1.

2.1. Feature Extraction

Each recording is resampled to 8 kHz, peak-normalized, and converted to a log-mel spectrogram $\mathbf{X} \in \mathbb{R}^{T \times F}$, where T denotes the number of time frames and $F = 128$ the number of mel bands. Spectrograms are computed using a 512-sample Hann window with an 80-sample hop, covering the 200–4000 Hz range.

Three preprocessing steps are subsequently applied to $\mathbf{X}$. First, amplitude values above the 99.5th percentile of each spectrogram are clipped to suppress sporadic impulsive noise. Second, with probability $p = 0.5$ sampled once per clip, a per-band noise floor $\hat{n}_f$, estimated

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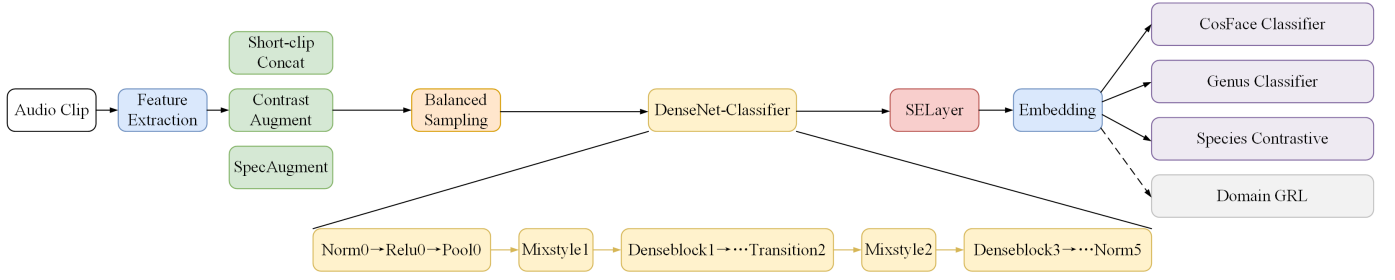


Fig. 1: Overview of the proposed framework. Log-mel features are augmented and class-balanced sampled before being fed into a DenseNet121 encoder with MixStyle and SE recalibration, followed by three task-specific classification heads (species, genus, domain) together with a supervised contrastive embedding objective. Dashed arrows indicate the reversed-gradient path through the GRL during back-propagation for domain-invariant representation learning.

as the 10th percentile along the time axis, is subtracted from each mel band f , reducing site-specific stationary background hum. Third, clips exceeding $L = 200$ frames (2 s) are cropped to a fixed-length window of L frames; the crop offset is sampled uniformly at random during training and fixed at the start of the clip during validation and testing.

2.2. Imbalance and Domain-Aware Augmentation

To address the class imbalance and domain-shift challenges introduced in Section 1, three training-only augmentations are applied sequentially to $\mathbf{X}$ after feature extraction.

Short-clip concatenation. Let n denote the number of frames in a given clip. If $n < 65$, the clip is concatenated along the time axis with a randomly sampled clip of the same species label, provided one is available; otherwise the clip is left unchanged. This mitigates the disproportionate representation of rare species caused by short recordings, which would otherwise both waste padding capacity and limit the temporal context available to the encoder.

Amplitude scaling. The spectrogram $\mathbf{X}$ is scaled by a factor $\alpha \sim \mathcal{U}(0.5, 1.5)$, simulating amplitude variation introduced by differing microphones, gain settings, and source distances across recording sites. This input-level perturbation complements the MixStyle and domain-adversarial mechanisms described in Section 2.4, which target domain invariance within the network.

SpecAugment. One time mask of up to 10 frames and one frequency mask of up to 16 mel bins are applied at randomly selected positions [11]. For the three most underrepresented species (*An. dirus*, *An. minimus*, *An. stephensi*; species indices 5, 7, and 8), the number of masks of each type is doubled, concentrating stronger regularization on the classes most prone to overfitting given their limited training samples.

2.3. Class-Balanced Sampling

Despite the augmentations above, the training set remains heavily imbalanced across the nine target species. Each training sample of species c is assigned a sampling weight following the effective-number formulation of [12]:

$$w_c = \frac{1 - \beta}{1 - \beta^{n_c}}, \quad \beta = 0.999, \quad (1)$$

where n_c denotes the number of training samples belonging to species c , and $\beta \in [0, 1)$ controls the rate at which the effective sample size saturates as n_c grows. The weight w_c is used both for weighted sampling during training and, as described in Section 2.5, for re-weighting the species classification loss.

2.4. Model Architecture

Backbone. The log-mel feature $\mathbf{X}$ is treated as a single-channel input of shape $(B, 1, T, F)$, where B is the batch size, and is processed by a DenseNet121 encoder [13] pretrained on ImageNet. The first convolutional layer, originally defined for three-channel RGB input, is adapted to single-channel input by summing its pretrained weights along the channel dimension, which preserves the magnitude of the original response. A batch normalization layer operating along the frequency axis is prepended to the backbone to standardize the input distribution.

Domain-invariant feature perturbation. To encourage domain-invariant representation learning, MixStyle modules [14] are inserted after the `pool0` and `transition2` stages of the backbone. For an intermediate feature map $\mathbf{x}$ and a randomly permuted sample $\mathbf{x}'$ within the same mini-batch, MixStyle computes

$$\hat{\mathbf{x}} = \sigma_{\text{mix}} \cdot \frac{\mathbf{x} - \mu(\mathbf{x})}{\sigma(\mathbf{x})} + \mu_{\text{mix}}, \quad (2)$$

where $\mu(\cdot)$ and $\sigma(\cdot)$ denote the instance-level channel-wise mean and standard deviation, and

$$\mu_{\text{mix}} = \lambda \mu(\mathbf{x}) + (1 - \lambda) \mu(\mathbf{x}'), \quad \sigma_{\text{mix}} = \lambda \sigma(\mathbf{x}) + (1 - \lambda) \sigma(\mathbf{x}'), \quad (3)$$

with mixing coefficient $\lambda \sim \text{Beta}(1, 1)$. By randomly interpolating feature statistics across samples, MixStyle simulates domain variation at the feature level during training.

Channel recalibration. Following the final dense block, a Squeeze-and-Excitation (SE) module [15] with channel reduction ratio 16 re-weights each output channel using a scalar gate derived from global average pooling.

Embedding projection. The pooled feature is projected to a 128-dimensional embedding $\mathbf{e} \in \mathbb{R}^{128}$ through a two-layer multilayer perceptron (Linear–BN–ReLU–Dropout(0.5)–Linear–BN), followed by L2 normalization. The embedding $\mathbf{e}$ is shared across three task-specific heads described below.

Species head. The species head applies a CosFace margin [16] to the cosine similarity between $\mathbf{e}$ and each class weight vector. Let θ_c denote the angle between the L2-normalized embedding and the L2-normalized weight vector of class c , and let y denote the ground-truth species label. The logit for class c is defined as

$$z_c = \begin{cases} s(\cos \theta_c - m), & c = y, \\ s \cos \theta_c, & c \neq y, \end{cases} \quad (4)$$

where $s = 30$ is a fixed scale factor and $m = 0.2$ is the angular margin, enlarging the decision boundary between species during training. The margin is disabled at inference time.

Genus head. A linear layer maps $\mathbf{e}$ to three genus logits (*Aedes*, *Culex*, *Anopheles*). This auxiliary task exploits phylogenetic structure shared among species of the same genus, providing an additional, taxonomically consistent supervisory signal.

Domain head. A two-layer MLP predicts the recording domain (D1–D5) from $\mathbf{e}$. A gradient reversal layer (GRL) [17] with coefficient $\alpha_{\text{grl}} = 1.0$ is inserted between the embedding and the domain classifier, negating the gradient during back-propagation so that the shared encoder is discouraged from retaining domain-identifying information. Since *Anopheles dirus* (species index 5) is the least-represented species and has not yet developed reliable species-discriminative features early in training, its samples are excluded from the domain loss via a binary mask:

$$\mathcal{L}_{\text{domain}} = \frac{\sum_{i=1}^B m_i \ell_i^d}{\sum_{i=1}^B m_i}, \quad m_i = \begin{cases} 1, & c_i \neq 5 \\ 0, & c_i = 5 \end{cases}, \quad (5)$$

where ℓ_i^d is the per-sample domain classification loss and $m_i \in \{0, 1\}$ indicates whether sample i contributes to $\mathcal{L}_{\text{domain}}$.

2.5. Multi-Task Training Objective

The overall training objective combines four loss terms:

$$\mathcal{L} = \alpha_s \mathcal{L}_{\text{species}} + w_g \mathcal{L}_{\text{genus}} + \lambda_d \mathcal{L}_{\text{domain}} + \alpha_c \mathcal{L}_{\text{contrast}}, \quad (6)$$

with fixed weights $\alpha_s = 1.0$, $\lambda_d = 0.5$, and $\alpha_c = 0.5$; the genus loss weight w_g is scheduled as described below.

Species loss. $\mathcal{L}_{\text{species}}$ is a focal loss [18] ($\gamma = 2.0$) with label smoothing ($\varepsilon = 0.1$), computed on the CosFace logits z_c from Eq. (4). Each sample’s contribution is additionally scaled by the effective-number weight w_c from Eq. (1).

Genus loss. $\mathcal{L}_{\text{genus}}$ is a cross-entropy loss with label smoothing ($\varepsilon = 0.1$) computed on the genus head. Its weight decays linearly over the first 50 epochs and is thereafter fixed:

$$w_g = \max \left(0.5, 1.0 - \frac{e - 1}{0.5T} \right), \quad (7)$$

where e is the current epoch and T is the total number of training epochs.

Domain loss. $\mathcal{L}_{\text{domain}}$ is the masked cross-entropy loss defined in Eq. (5), computed on the domain head after gradient reversal.

Supervised contrastive loss. Following [19], a supervised contrastive loss with temperature $\tau=0.05$ is applied on the L2-normalised embeddings. It treats all same-species samples as positives, pulling same-species embeddings together and pushing different-species embeddings apart in hyperspherical space.

3. EXPERIMENTAL SETUP

3.1. Dataset

We use the BioDCASE 2026 Task 5 development dataset, which comprises nine species across five recording domains (D1–D5): *Aedes aegypti*, *Aedes albopictus*, *Culex quinquefasciatus*, *Anopheles gambiae*, *Anopheles arabiensis*, *Anopheles dirus*, *Culex pipiens*, *Anopheles minimus*, and *Anopheles stephensi*. We follow the official training/validation (trainval) and test splits.

Table 1 shows the number of test samples from the unseen domain for each species, where the unseen domain indicates the recording location never seen during training for that species.

Table 1: Test samples from the unseen domain per species.

Species	Unseen Domain	Unseen Domain Samples
<i>Ae. aeg</i>	D3	192
<i>Ae. alb</i>	D2	419
<i>Cx. qui</i>	D1	672
<i>An. gam</i>	D1	818
<i>An. ara</i>	D1	1 820
<i>An. dir</i>	D4	40
<i>Cx. pip</i>	D3	68
<i>An. min</i>	D2	99
<i>An. ste</i>	D4	74

Table 2: Cross-domain performance of different CNN backbones (baseline setup). Values are balanced accuracy (BA) and Domain Shift Gap (DSG) on test split, mean $\pm$ std over ten seeds.

Backbone	BA _{seen}	BA _{unseen}	DSG
MTRCNN	0.881 $\pm$ 0.011	0.175 $\pm$ 0.020	0.706 $\pm$ 0.025
ResNet	0.903 $\pm$ 0.005	0.171 $\pm$ 0.035	0.732 $\pm$ 0.036
Inception V3	0.863 $\pm$ 0.009	0.187 $\pm$ 0.013	0.676 $\pm$ 0.015
MobileNet	0.826 $\pm$ 0.232	0.134 $\pm$ 0.030	0.692 $\pm$ 0.226
Xception	0.888 $\pm$ 0.015	0.149 $\pm$ 0.032	0.739 $\pm$ 0.033
EfficientNet	0.902 $\pm$ 0.008	0.113 $\pm$ 0.034	0.789 $\pm$ 0.029
DenseNet121	0.912 $\pm$ 0.008	0.194 $\pm$ 0.017	0.718 $\pm$ 0.019

3.2. Training

We use the AdamW optimizer [20] with a learning rate of 10^{-3} and weight decay of 10^{-4} . The batch size is set to 64, and the maximum training epoch is 100. ReduceLROnPlateau scheduler decays the learning rate by a factor of 0.5 if the validation species balanced accuracy shows no improvement for 6 consecutive epochs, with a lower bound of 10^{-6} . Early stopping is adopted with a warm-up of at least 15 epochs and a patience of 10. All training experiments are performed on a single NVIDIA RTX 3090 GPU.

3.3. Metrics

Following the challenge protocol, we report species-level balanced accuracy (macro-average recall) separately for seen domains (BA_{seen}) and unseen domains (BA_{unseen}), as well as per-domain BA. The official ranking is primarily determined by BA_{unseen}, with the Domain Shift Gap (DSG) defined as the absolute difference between BA_{unseen} and BA_{seen} used as a secondary metric (lower is better).

4. RESULTS AND DISCUSSION

4.1. Backbone Architecture Comparison

We first compare seven CNN backbones (ResNet [21], InceptionV3 [22], MobileNet [23], Xception [24], EfficientNet [25], DenseNet [13], and MTRCNN [26]) under a baseline setup (cross-entropy loss only, without class balancing or domain adaptation) to identify a suitable feature extractor. As shown in Table 2, DenseNet121 achieves the highest BA on both seen and unseen domains. Given that the challenge primarily emphasizes unseen-domain performance, DenseNet121 is adopted as the backbone for our best-performing setup.

4.2. Progressive Module Ablation Study

We conduct a progressive ablation experiment based on MTRCNN baseline to verify the incremental gain of each proposed imbalance and domain adaptation component. We add modules one by one in the order: balanced sampling, targeted SpecAugment, selective GRL domain-adversarial training, and short-clip concatenation. Results averaged over random seeds are listed in Table 5.

Table 3: Ablation study of loss components on the test set.

$\mathcal{L}_{\text{species}}$	$\mathcal{L}_{\text{genus}}$	$\mathcal{L}_{\text{domain}}$	$\mathcal{L}_{\text{contrast}}$	BA _{seen}	BA _{unseen}	DSG
✓	✓	—	—	0.892 ± 0.006	0.288 ± 0.008	0.604 ± 0.010
✓	✓	—	✓	0.889 ± 0.008	0.266 ± 0.023	0.623 ± 0.017
✓	✓	✓	—	0.858 ± 0.021	0.256 ± 0.025	0.602 ± 0.018
✓	✓	✓	✓	0.875 ± 0.018	0.299 ± 0.014	0.576 ± 0.008

Table 4: Unseen-domain per-species recall of individual ensemble component models.

Model	Encoder	n_{mels}	Freq. mask	<i>Ae. aeg</i>	<i>Ae. alb</i>	<i>Cx. qui</i>	<i>An. gam</i>	<i>An. ara</i>	<i>An. dir</i>	<i>Cx. pip</i>	<i>An. min</i>	<i>An. ste</i>
Model 1	DenseNet121	128	16	0.3958	0.5251	0.0000	0.0098	0.0044	0.0000	0.9265	0.8586	0.4595
Model 2	DenseNet121	64	4	0.0000	0.4248	0.0030	0.2934	0.0742	0.0000	0.8824	0.8283	0.5946
Model 3	MTRCNN	64	2	0.0000	0.3007	0.0074	0.0122	0.0962	0.0000	0.8382	0.6061	0.6892

Table 5: Progressive ablation of core proposed modules based on MTRCNN baseline. Metrics are BA_{seen}, BA_{unseen} and DSG on test split, mean ± std over ten training seeds.

Module	BA _{seen}	BA _{unseen}	DSG
Baseline	0.881 ± 0.011	0.175 ± 0.020	0.706 ± 0.025
+ Balanced Samp.	0.871 ± 0.015	0.238 ± 0.011	0.634 ± 0.021
+ SpecAug.	0.858 ± 0.033	0.245 ± 0.016	0.612 ± 0.032
+ Selective GRL	0.827 ± 0.025	0.266 ± 0.018	0.561 ± 0.037
+ Short-clip Concat.	0.821 ± 0.013	0.275 ± 0.025	0.546 ± 0.030

4.3. Effect of Multi-Task Learning Objectives

We perform controlled-variable ablations within the full framework in Sec. 2, fixing all modules and varying only the loss combinations to isolate each loss’s contribution. Results averaged over five random seeds are shown in Table 3.

Adding contrastive or domain-adversarial loss alone fails to improve unseen-domain performance. By contrast, the full combination yields the best unseen BA and lowest DSG, confirming that joint optimisation of the four losses effectively mitigates domain shift and class imbalance.

4.4. Ensemble Enhancement

As a baseline reference prior to introducing our final system, we evaluate the best single model, which achieves BA_{unseen} = 0.3533 (Table 6). We then combine three complementary models via confidence-gated cascade fusion. The components use different backbones, mel resolutions, and SpecAugment configurations (Table 4). Fusion proceeds in three rounds:

- 1) *Round 1 (high precision)*: The primary model’s prediction is accepted only if its softmax confidence exceeds a species-specific threshold (0.5–0.6), retaining only highly trustworthy decisions.
- 2) *Round 2 (balanced)*: Samples rejected by Round 1 are passed to a second model. Its prediction is adopted when confidence meets a lower threshold, striking a compromise between precision and recall.
- 3) *Round 3 (full coverage)*: Samples still unresolved after Rounds 1 and 2—specifically those from species with poor per-model performance (e.g., *Anopheles dirus*)—are handled by a third model without any confidence constraint, ensuring a prediction for every test clip.

Table 7 reports the ensemble performance. Compared with the single model, the ensemble improves overall test BA from 0.5159 to 0.5602, unseen-domain BA from 0.3533 to 0.4016, and species accuracy from 0.6219 to 0.6361, while reducing DSG from 0.3228 to 0.2987.

Table 6: Single-model performance on validation and test splits.

Metric	Validation	Test
Species Accuracy	0.7051	0.6219
Species BA (overall)	0.7495	0.5159
BA _{D1}	0.8512	0.3337
BA _{D2}	1.0000	0.7390
BA _{D3}	0.7778	0.6074
BA _{D4}	0.5000	0.1149
BA _{D5}	0.6740	0.6762
BA _{seen}	—	0.6761
BA _{unseen}	—	0.3533
DSG	—	0.3228

Table 7: Ensemble system performance on validation and test splits.

Metric	Validation	Test
Species Accuracy	0.7107	0.6361
Species BA (overall)	0.7457	0.5602
BA _{D1}	0.7887	0.4221
BA _{D2}	0.7216	0.7135
BA _{D3}	0.7302	0.5886
BA _{D4}	0.5000	0.1182
BA _{D5}	0.6990	0.7003
BA _{seen}	—	0.7003
BA _{unseen}	—	0.4016
DSG	—	0.2987

4.5. Results on the Official Evaluation Dataset

On the official evaluation dataset, the ensemble model achieves BA_{unseen} = 0.205, BA_{seen} = 0.636, and DSG = 0.431. This performance still surpasses the official baseline (BA_{unseen} : 0.172, BA_{seen} : 0.582, DSG : 0.409), yet the improvement is notably smaller than that observed on the development test set.

5. CONCLUSION

We propose a framework for cross-domain mosquito species classification to tackle class imbalance and domain shift by integrating data augmentation, MixStyle modules, multi-task optimization, and confidence-gated ensemble fusion. Although our approach achieves substantial improvements over the official baseline on the development test set and demonstrates the benefits of joint multi-task training, these gains cannot be fully reproduced on the official evaluation dataset. The limited performance improvement on unseen domains indicates insufficient generalization to entirely unseen recording conditions. For future work, we aim to explore more generalizable domain-adaptation methods and robust ensemble strategies to reduce the performance gap between development and evaluation domains.

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